

PORT ALSWORTH 1 SW, AK, USA

WMO: 703490

Lat: 60.1951N

Lon: 154.3196W

Elev: 98

StdP: 100.16

Time Zone: -9.00 (NAK)

Period: 09-19

WBAN: 26562

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-27.4	-34.7	0.1	-31.7	-30.1	0.2	-26.8	3.9	2.6	3.6	4.5	0.0	N/A	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	24.7	15.4	22.3	14.2	20.2	13.4	16.3	23.1	15.2	20.8	14.1	18.8	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	9.8	17.9	12.6	9.2	16.7	11.8	8.7	15.5	45.6	23.1	42.5	20.7	39.8	19.1	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.1	2.7	2.2	DB	-31.6	28.3	5.3	2.8	-35.4	30.3	-38.5	31.9	-41.4	33.5	-45.3	35.5	
			WB	-31.7	17.4	5.2	1.6	-35.5	18.6	-38.6	19.5	-41.5	20.4	-45.3	21.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-8.0	-5.3	-5.7	1.9	7.2	11.7	13.7	12.5	8.3	2.7	-4.9	-7.4
	DBStd	10.09	11.10	8.09	7.68	4.13	3.09	2.64	2.69	2.41	3.18	4.82	7.25	9.22
	HDD10.0	3114	557	430	486	243	97	12	2	6	67	231	446	539
	HDD18.3	5871	816	663	744	493	346	201	146	182	301	486	696	798
	CDD10.0	288	0	0	0	0	9	62	117	82	16	3	0	0
	CDD18.3	3	0	0	0	0	0	1	3	0	0	0	0	0
	CDH23.3	135	0	0	0	0	5	55	71	4	0	0	0	0
	CDH26.7	29	0	0	0	0	0	13	16	0	0	0	0	0
Wind	WSAvg	0.8	0.8	0.9	0.9	1.0	1.0	0.9	0.7	0.6	0.7	0.8	0.8	0.8
Precipitation	PrecAvg	399	25	25	19	17	13	19	38	57	68	43	59	33
	PrecMax	619	72	68	52	43	35	43	109	108	170	92	364	80
	PrecMin	259	5	3	2	4	3	4	4	20	11	10	6	7
	PrecStd	97	18	18	11	10	9	10	23	27	37	24	74	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	8.9	10.8	13.8	23.6	28.8	29.2	23.6	19.8	15.3	10.4	8.1
		MCWB	5.9	4.6	5.0	6.1	12.6	15.8	18.7	15.4	13.0	8.6	5.6	3.6
	2%	DB	8.2	6.9	8.3	12.0	20.5	24.6	25.1	21.5	17.0	13.0	9.0	7.0
		MCWB	4.1	2.8	3.7	5.4	10.8	14.6	16.4	14.7	11.3	7.3	4.8	3.0
	5%	DB	6.8	5.6	6.4	10.6	17.1	21.5	22.7	19.5	15.2	11.5	7.1	5.8
		MCWB	3.2	2.0	2.7	4.7	9.2	13.2	15.1	13.6	10.4	6.6	4.1	2.4
	10%	DB	5.6	4.5	4.9	9.2	14.6	18.9	20.4	17.8	13.7	9.9	5.2	4.3
		MCWB	2.5	1.3	1.5	4.1	8.0	11.9	14.1	13.0	9.9	6.1	2.7	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	4.6	5.4	6.9	13.0	16.8	18.9	16.4	13.3	8.9	6.5	4.1
		MCDB	11.2	7.7	9.9	12.4	21.9	26.3	28.4	21.7	18.1	13.0	9.6	7.0
	2%	WB	4.5	3.0	4.1	5.7	11.2	15.1	17.0	15.3	12.2	7.8	5.3	3.4
		MCDB	7.8	6.1	8.1	10.9	19.2	23.7	24.3	19.6	15.4	11.6	8.2	6.2
	5%	WB	3.5	2.2	3.1	5.1	9.6	13.7	15.7	14.5	11.4	7.1	4.3	2.7
		MCDB	6.4	5.1	6.3	10.0	16.3	20.6	21.8	18.4	14.1	10.4	6.9	5.4
	10%	WB	2.5	1.5	1.8	4.3	8.4	12.5	14.7	13.6	10.5	6.3	3.1	1.7
		MCDB	5.4	4.1	4.6	8.6	13.9	18.2	19.8	17.0	13.0	9.2	5.5	4.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.4	12.3	11.5	12.8	12.7	10.3	9.0	9.4	8.2	7.6	7.6
		MCDBR	7.2	7.6	10.7	14.0	19.1	18.9	16.2	13.1	11.6	9.5	7.6	6.7
		MCWBR	4.5	4.9	6.8	8.3	10.1	9.3	7.9	6.8	6.8	5.9	4.9	4.4
	5% WB	MCDBR	7.2	7.3	9.6	12.3	17.3	16.8	15.0	11.2	10.3	8.1	7.0	6.5
		MCWBR	4.6	4.9	6.1	7.3	9.4	8.6	7.5	6.0	6.5	5.2	4.8	4.4
Clear-Sky Solar Irradiance	taub	0.240	0.276	0.300	0.390	0.383	0.392	0.370	0.361	0.327	0.299	0.254	0.235	
	taud	2.096	2.214	2.183	1.970	2.167	2.273	2.413	2.424	2.495	2.350	2.060	1.929	
	Ebn at Noon	526	725	824	790	837	837	851	830	804	694	479	335	
	Edh at Noon	43	69	98	148	131	121	103	94	74	60	43	32	
All-Sky Solar Radiation	RadAvg	0.36	1.03	2.74	4.27	4.93	5.24	4.40	3.48	2.37	1.17	0.47	0.20	
	RadStd	0.05	0.16	0.36	0.45	0.56	0.44	0.41	0.46	0.35	0.12	0.08	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.83	+2.66	+2.71	N/A	N/A	N/A	N/A	N/A	-304	N/A	N/A

Nomenclature: See separate page